

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

- CampusMesh** Jan 2026 – Present
Software Engineer
New York, US
 - Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
 - Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda**, **API Gateway**, **S3**, and **CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys**, **SSM Parameter Store**, and **least-privilege IAM**.
 - Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
Seattle, US
 - Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented **system design** for approval and established Role-Based Access (**4+ roles**) via **OAuth**, **Midway SSO**, **HELIS**.
 - Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, IAM, and **CloudWatch**; reduced provisioning time **70%** with **multi-stage, multi-region CI/CD**.
 - Built a **React 18** frontend with optimized API integration, advanced filtering, and client-side caching, reducing page load time by **40%**.
 - Engineered backend services in **Java** (Coral, Smithy, Dagger, CBOR); triaged defects using **application logs** and metrics, fixed bugs with disciplined **unit and integration tests** via **Mockito**, cutting backend execution time by **35%**.
- New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
Software Developer
New York, US
 - Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
 - Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards for operational review.
 - Developed an insurance analytics platform (**React**, **Node.js**, **REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue across **6 clients**.
 - Documented issue findings and resolution paths for cross-functional partners; implemented **WebSocket-based real-time communication** and broker-facing workflows integrated with insurer platforms.
- PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
Bengaluru, India
 - Developed a distributed backend on **Microsoft Azure** for a **health platform** integrating with **Microsoft CRM**, diagnosing production issues by correlating **application logs**, SQL traces, and operational data across services.
 - Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
 - Optimized data ingestion with **parallel processing** on **100K-row** Excel feeds, cutting execution from **5 hours to 1 hour**.

Projects

- InviLite (Incident Analysis & Data Platform)** | *Python, AWS, Kinesis, Lambda, DynamoDB, Bedrock, FAISS*
 - Built a cloud-native pipeline (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured operational data for governed analysis workflows.
 - Developed a **RAG system** over historical incident records and runtime signals, accelerating root-cause investigation from hours to minutes and lifting classification accuracy from **56% to 92%**.
- Hospital Management System (Full-Stack Application)** | *React, Node.js, Express, MongoDB, Jest, AWS SageMaker*
 - Built a **full-stack healthcare platform** (React, Node.js, Express, MongoDB) managing appointments, patient records, lab reports, and medical history with **role-based access control**.
 - Wrote **automated tests** for API and UI flows; fine-tuned a **Hugging Face transformer-based model** reaching **88% accuracy** for health condition prediction on **AWS SageMaker**.
- Advanced Expense Tracker (Python Desktop Application)** | *Python, Tkinter, SQLite3, Pandas, Matplotlib*
 - Architected a modular **Python application** for individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns** over **SQLite3**.
 - Implemented concurrent transaction handling and real-time cost splitting using **Python threading** and locks for thread-safe DB writes; visualized spending trends with **Matplotlib**.

Skills

- Programming Languages:** C++, Java, Python, C#, TypeScript, JavaScript (ES6), SQL
- Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices, HTML, CSS, Sequelize
- Databases:** MySQL, PostgreSQL, DynamoDB, MongoDB, Firebase, Dataverse
- Cloud & DevOps:** AWS (CDK, EC2, S3, DynamoDB, API Gateway, Lambda, CloudWatch, Cognito, IAM, SageMaker), Azure (OpenAI, Function Apps, Storage), Docker, Kubernetes, CI/CD
- AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, Hugging Face
- Tools & Platforms:** Git, Linux, Visual Studio, Jest, Mockito, Postman, Jupyter, Power BI, QuickSight

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals